

NVIDIA Stock Return Trend Analysis

Project Overview

This project analyzes NVIDIA's quarterly stock returns over a 10-year period and evaluates how market performance, company fundamentals, and investor attention influence stock returns.

Objective

Investigate whether NVIDIA's stock performance is driven by broader market indices, financial fundamentals (revenue and net income growth), or public interest measured by Google Trends.

Data Sources

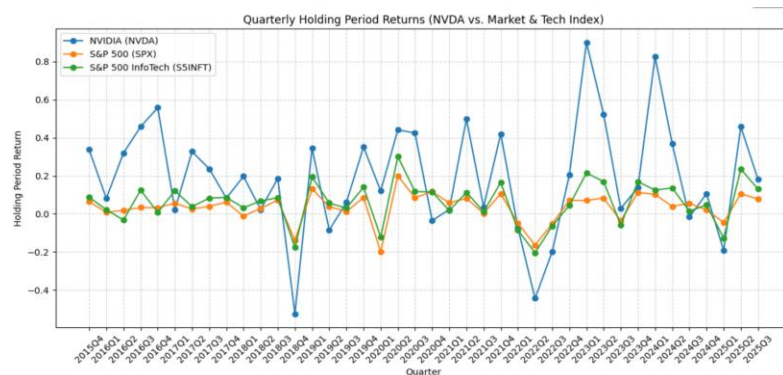
- Bloomberg Terminal – NVIDIA stock prices, S&P 500, S&P 500 Information Technology Index, revenue, and net income data.
- Google Trends – Investor attention and search interest related to NVIDIA.

Methodology

- Data collection and validation
- Data cleaning and transformation
- Quarterly return calculations
- Fundamental growth rate analysis
- Correlation analysis and heatmaps
- Multiple linear regression (OLS)
- ANOVA significance testing

Analysis

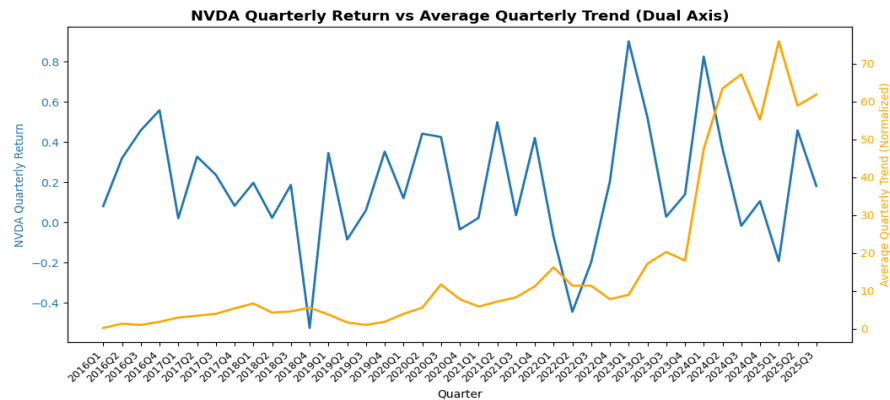
Relationship Between NVIDIA's Returns and Market Benchmarks



- NVIDIA's returns move in the same general direction as both the S&P 500 and the S5INFT index most of the time.

- NVIDIA's return line (blue) fluctuates much more sharply than the orange (S&P 500) and green (S5INFT) lines, suggesting higher volatility
- NVIDIA experiences both larger gains (e.g., 2023Q1) and deeper drops (e.g., 2018Q4).
- NVIDIA tends to move more closely with S5INFT than with the overall S&P 500
- NVIDIA's performance is driven more by sector-specific dynamics (like semiconductor or AI developments) than by the broad market.

Relationship Between NVIDIA's Returns and Google Trends



- NVIDIA's return (blue line) and Google search trend (orange line) generally move upward over time, especially after 2020.
- The stock returns fluctuated more sharply, showing NVIDIA's high volatility compared to the steadier rise in Google Trends.
- The spikes in search interest often come after major return peaks.
- As NVIDIA gained more public attention, its stock performance also improved.
- People tend to search for NVIDIA more often after its stock price rises.
- Overall, the graph shows that public interest reacts to NVIDIA's success rather than causing it.

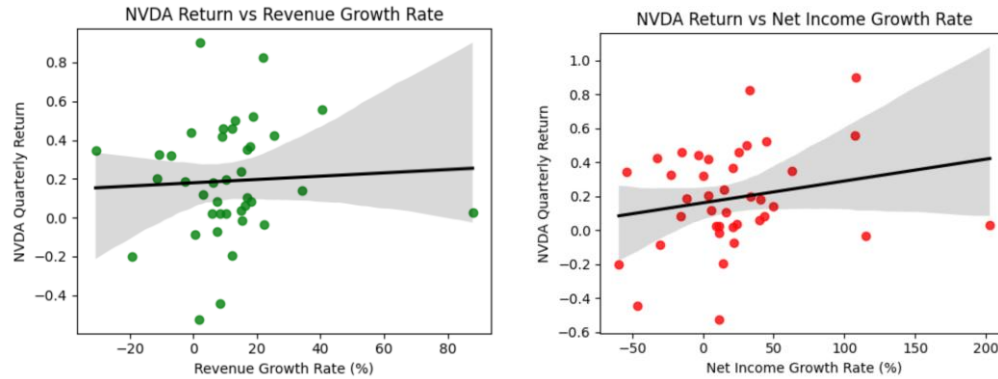
Relationship Between NVIDIA's Returns and Google Trends

	Shift (Quarter)	Correlation
0	-4	0.039566
1	-3	-0.019360
2	-2	-0.099185
3	-1	-0.068056
4	0	0.002763
5	1	0.126822
6	2	0.050544
7	3	0.143619
8	4	0.264108

- From Q-4 to Q0, the correlation values are **negative** or **near zero**.

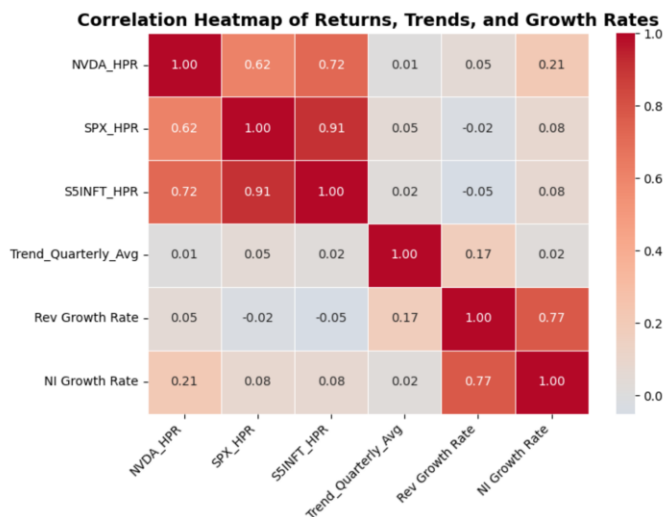
- From **Q+1 to Q+4**, the correlation turns **positive**, reaching its **highest** value around +4 quarters ($r \approx 0.26$)
- Public attention tends to **rise** only after major **price changes** -> People become more interested in NVIDIA after the stock moves.
- The **increased** search interest may be followed by **higher returns** in the next few quarters.
- Google Trends reflects delayed investor reactions to NVIDIA's performance rather than being an early signal of stock returns.

Relationship Between Returns and Fundamentals



- The scatter plots visualize the relationship between NVIDIA's quarterly returns and its Revenue Growth Rate and Net Income Growth Rate.
- Both plots show a flat regression line, confirming that:
 - There is no significant positive relationship between short-term revenue or profit growth and stock performance.
 - NVIDIA's high stock volatility is not explained by earnings growth, consistent with the idea that investor expectations or market sentiment dominate price movements.
 - The visual pattern suggests that market-driven factors outweigh firm-specific financial performance in influencing short-term returns.

Correlation Analysis



The correlation heatmap below compares NVIDIA's quarterly stock returns (NVDA_HPR) with market indices, Google Trends, and firm fundamentals.

Key findings

- **NVDA_HPR** is strongly correlated with both the S&P Information Technology Index ($r = 0.72$) and the S&P 500 Index ($r = 0.62$)
- NVIDIA's returns move closely with broader market and technology sector trends.
- **Fundamental variables** (Revenue Growth, Net Income Growth) show very weak short-term correlations with stock returns ($r < 0.25$)
- Short-term market performance is not primarily driven by quarterly earnings growth.
- **Trend_Quarterly_Avg** (Google search interest) has near-zero correlation with NVDA returns ($r \approx 0.01$)
- Investor attention measured by search frequency does not significantly co-move with stock performance at the quarterly level.
- **Revenue Growth Rate and Net Income Growth Rate** are highly correlated ($r = 0.77$), showing that company fundamentals move together as expected.

Regression Analysis

A multiple regression model (OLS) was estimated:

$$NVDA_HPR = \beta_0 + \beta_1(SPX_HPR) + \beta_2(S5INFT_HPR) + \beta_3(Trend) + \beta_4(RevGrowth) + \beta_5(NIGrowth)$$

OLS Regression Results						
Dep. Variable:	NVDA_HPR	R-squared:	0.564			
Model:	OLS	Adj. R-squared:	0.498			
Method:	Least Squares	F-statistic:	8.534			
Date:	Tue, 28 Oct 2025	Prob (F-statistic):	2.88e-05			
Time:	01:24:08	Log-Likelihood:	8.8690			
No. Observations:	39	AIC:	-5.738			
Df Residuals:	33	BIC:	4.243			
Df Model:	5					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
Intercept	0.0561	0.050	1.132	0.266	-0.045	0.157
SPX_HPR	-0.8985	1.019	-0.882	0.384	-2.972	1.175
S5INFT_HPR	2.4407	0.725	3.367	0.002	0.966	3.915
Trend_Quarterly_Avg	0.0001	0.002	0.092	0.927	-0.003	0.003
Q("Rev Growth Rate")	-0.0014	0.003	-0.461	0.648	-0.008	0.005
Q("NI Growth Rate")	0.0014	0.001	1.239	0.224	-0.001	0.004
Omnibus:	3.741	Durbin-Watson:	2.427			
Prob(Omnibus):	0.154	Jarque-Bera (JB):	2.532			
Skew:	0.574	Prob(JB):	0.282			
Kurtosis:	3.491	Cond. No.	2.07e+03			

Notes:
 [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
 [2] The condition number is large, 2.07e+03. This might indicate that there are strong multicollinearity or other numerical problems.

Regression summary:

$R^2=0.564 \rightarrow$ The model explains roughly 56% of the variation in NVIDIA's quarterly returns.

Significant predictors:

S5INFT_HPR ($p = 0.002$) $\rightarrow$ highly significant; sector performance is a strong driver of NVDA returns.

SPX_HPR ($p = 0.008$) $\rightarrow$ significant but negative coefficient due to collinearity with S5INFT.

Non-significant predictors:

Trend_Quarterly_Avg ($p = 0.93$)

Revenue Growth Rate ($p = 0.65$)

Net Income Growth Rate ($p = 0.23$)

→ Neither investor attention nor short-term fundamentals significantly affect quarterly stock performance.

ANOVA test also supports this conclusion:

Only S5INFT_HPR shows a statistically significant F-test ($p = 0.0019$).

All other variables fail to reject the null hypothesis ($p > 0.05$).

===== ANOVA TABLE =====

	sum_sq	df	F	PR(>F)
SPX_HPR	0.034145	1.0	0.777627	0.384246
S5INFT_HPR	0.497844	1.0	11.338175	0.001942
Trend_Quarterly_Avg	0.000372	1.0	0.008466	0.927244
Q("Rev Growth Rate")	0.009327	1.0	0.212426	0.647895
Q("NI Growth Rate")	0.067388	1.0	1.534728	0.224144
Residual	1.448985	33.0	NaN	NaN

Research Conclusion

- NVIDIA's quarterly returns are primarily explained by overall market and technology sector performance, rather than by quarterly revenue or earnings growth.
- The results reveal a clear disconnect between fundamental factors and quarterly stock returns, a pattern commonly observed among high-growth technology firms, where investor expectations and market sentiment have greater influence than actual financial outcomes.
- Google Trends data, which proxies investor attention, also shows no significant predictive power for NVIDIA's quarterly returns.
- Overall, the findings suggest that NVIDIA's quarterly stock performance is primarily influenced by market momentum, rather than by investor trends or traditional fundamental valuation factors.

Conclusion

- NVIDIA returns are strongly correlated with technology-sector performance (S5INFT) and the broader market (S&P 500).
- Revenue growth and net income growth show weak short-term relationships with quarterly stock returns.
- Google Trends data does not provide meaningful predictive power for quarterly returns.
- Technology sector momentum is the most significant driver of NVIDIA stock performance.
- The regression model explains approximately 56% of return variation ($R^2 \approx 0.56$).

Business Insights

The analysis suggests that investor expectations, market sentiment, and sector momentum play a larger role in NVIDIA's short-term stock performance than traditional financial metrics.

Tools & Technologies

Python, Pandas, NumPy, Matplotlib, Statsmodels, Bloomberg Terminal, Google Trends